

Inigo Quilez :: articles :: bitmap orbit traps - 2002



Back in 2002 I thought to extend the idea of **orbit traps** for coloring fractals so I wasn't restricted only to geometrical figures. My first idea was to "draw" the trapping area into a bitmap, and lookup in the bitmap for every point of the orbit to see if there was a "hit" or not.

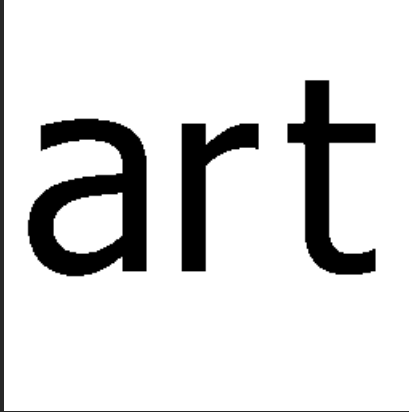


Coloring with bitmap orbit traps, made in 2002 - Click image to enlarge

The image above made it into the first position of the National Fractal Art Contest of 2002 in Spain (click [here](#) for the results). The pink-bunny image belongs to Made, the demoscene graphician.

The second idea was to do orbit traps with real distances to the trap. For that to work I made a black and white image with the trap area in black, and then for every white pixel I computed the distance to the closest black pixel. The resulting grey-scale image encoded the 2D distance field that I could use as true replacement for

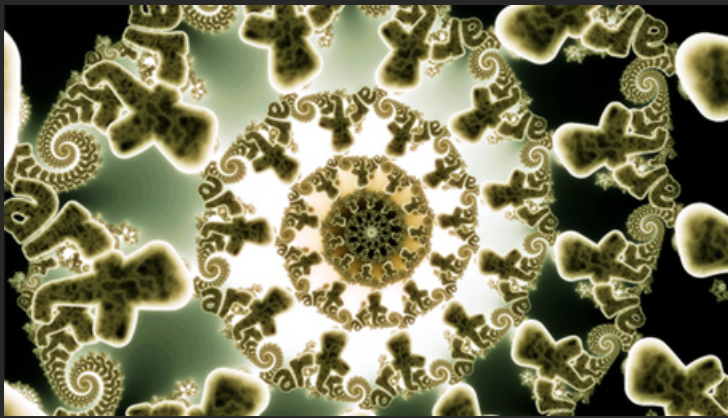
geometric traps in the regular orbit trap algorithm:



The input trap image



Precomputed distance field



Resulting image

An example of basic in/out trap algorithm can be found here: <https://www.shadertoy.com/view/4slGWH>



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